

If I Share with you my Perspective, Would you Share your Data with me?

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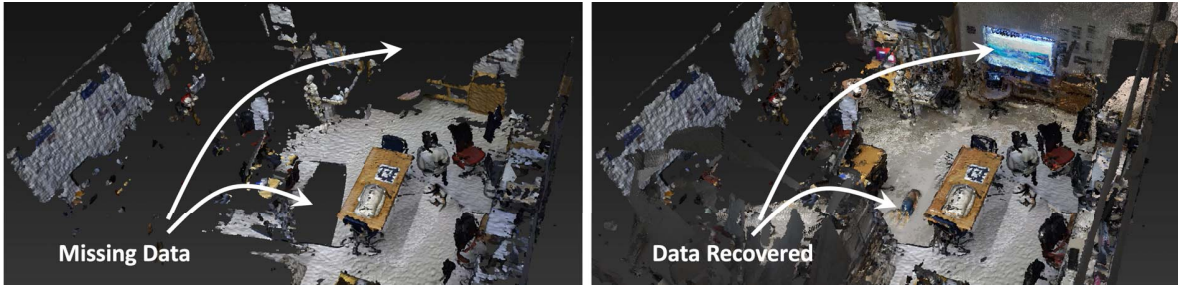


Figure 1: Thanks to dynamic calibration and data fusion, as the users wearing HMD enter and move around the environment, their Dynamic 3D views complement the reconstructed field of view of the room RGBD sensors.

ABSTRACT

Real-time 3D reconstruction using multiple RGBD cameras and their online transmission facilitates the adoption of mixed reality telepresence. However, such a system can only cover a limited volume, and increasing the number of RGBD cameras is unfeasible due to setup complexity and space constraints. To address this issue, we present the concept of Dynamic 3D View Sharing, which complements the views of a 3D reconstruction system by the dynamic view of the user's HMD. Here, we present a markerless calibration method integrating these two seamlessly into the mixed reality telepresence systems without disrupting the current workflow.

Index Terms: Human-centered computing—Mixed / augmented reality; Computing methodologies—Artificial intelligence—Computer vision—Reconstruction

1 INTRODUCTION

Digital reconstruction of the real world using a multi-camera setup has always been an active research direction. Such system usually consists of multiple RGBD sensors that are capable of capturing depth and texture information to create a 2.5D representation of the scene. The dynamically reconstructed scene can be recorded for playback later or streamed in real time to remote users on a head-mounted displays (HMD) [2]. However, there are still many challenges with multi-camera setups. As a result of the inherent characteristics of the sensing technologies, for example, reflective objects at a certain angle or darker objects at long distances are barely visible or appear noisy in the ToF depth camera. Placing additional cameras to alleviate these issues specifically is not desirable as it causes increased costs of equipment and labor.

Currently, the setup of mixed reality telepresence systems is still quite cumbersome and often requires an on-site expert to conduct extrinsic calibration to align the live point cloud streams captured by the multi-view camera setups. Moreover, the co-localization of the HMD to the multi-camera capturing system remains an issue every time a new user wearing HMD enters and leaves the scene.

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Our goal is to create a smoother telepresence experience such that once users with HMD walk into a scene, which is being reconstructed by a 3D capturing system, their inside-out 3D views are automatically registered to the digital world reconstructed by the existing system. Furthermore, we want to allow remote users to request additional scene information that could not be captured by the static setup. This work introduces the concept of Dynamic 3D View Sharing and proposes a set of building blocks to address the depth sensing and calibration issues of telepresence systems.

2 METHODS

HMD Perspective Sharing. We fetched timestamped 320×288 depth image from the ToF sensor on HoloLens 2 as well as the 2D to 3D mapping between the depth images coordinates and camera reference frame coordinate through Research Mode [3]. In addition, The timestamped 760×428 color image from the RGB camera and its intrinsic parameters as well as frame metadata containing the transformation between RGB camera and world coordinate, and the device to world coordinate system can be retrieved in runtime. We then used ZeroMQ¹ to receive frame data on a Ubuntu 18.04 workstation. The closest timestamp matching method was used to synchronize the color and depth frames on the server. The generated point cloud was transformed into the world space coordinate system, then projected into the corresponding color image to associate the color value for each point.

Markerless registration. The registration problem of HMD to the 3D capturing and reconstruction system can be formulated as follows:

$${}^cT_{\text{HMD}}(t_i) = {}^cT_w {}^wT_{\text{HMD}}(t_i) \quad (1)$$

where ${}^cT_{\text{HMD}}(t_i)$ is the pose of the HMD with respect to the coordinate system of the 3D capturing array, ${}^wT_{\text{HMD}}(t_i)$ is the transformation estimated by the inside-out tracking algorithm integrated in the HMD, and cT_w needs to be computed as described in the following. In order to establish sufficient feature correspondences between the HMD's perspective and the capturing system's perspective, we collected 40 frames of point cloud to cover approximately the same or more regions of the room. As the depth map produced from the ToF sensor on HoloLens 2 is noisy, we downsampled the point cloud with a voxel of 0.01 and filtered the point cloud by removing points that have fewer than 16 point neighbors in a given

¹ZeroMQ: <https://zeromq.org>

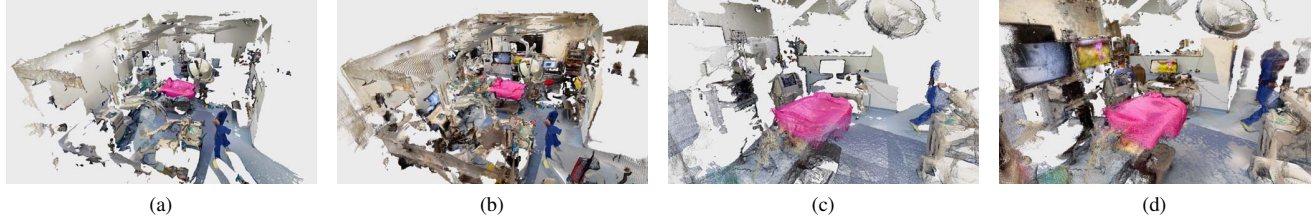


Figure 2: (a) The setup of the 3D capturing system in the operation room focuses mostly on the surgical table. However, a certain amount of information is lost. Perspective sharing from the local HMD user in the scene promotes capturing much richer content. As shown in (b), endoscopic and surgical monitor is revealed in the scene. (d) An alternative viewpoint of the scene reveals the information that are lost in (c).

0.05m sphere around them. We established correspondences by nearest neighbor search in feature space of the point cloud from both system and fed into the global point cloud registration method Teaser++ [4] for the initial calibration. We then took advantage of Colored ICP algorithm [1] to fine-tune the calibration result. The server, in the end, published the transformation matrix cT_w back to the HMD.

3 EXPERIMENT AND RESULTS

To showcase the use in a medical setting, we mounted four Azure Kinect sensors on the ceiling in an operating room. The cameras were pre-calibrated using a two-step approach: initial camera pose estimation with a fiducial marker and then refinement using Colored ICP. An HMD user entered the scene and walked around. The measured end-to-end latency of data streaming to the server is 157 ± 23 ms. The calibration process took less than 1 minute and the final measured live reconstruction frame rate is 29.97fps.

3.1 Perspective and Data Sharing Visualization

As shown in Fig. 2, although the 3D capturing system was able to reconstruct most of the scene, there are still many gaps within the fused point cloud. The point cloud is completed using the data from the HMD, revealing the monitors and steel cabinets that are hard to pick up by the ToF sensor, together with devices and equipment outside the reconstruction volume or occluded by other objects.

3.2 Accuracy Evaluation

By using the 3D feature points in the marker, we evaluated the accuracy of the system. The registration of the 3D capturing system had an overall error of 0.90 ± 0.51 cm. The error of HMD scene reconstruction was found to be 2.51 ± 2.92 cm, and the final registration from HMD to the kinect system had an average error of 2.95cm.

4 GENERAL DISCUSSION AND CONCLUSION

We discussed the integration of one exemplary 3D dynamic view of an incoming user, however the system could easily be extended to integrating multiple HMD users and enable dynamic updating of the world's digital model using all available perspectives of dynamic users who wish to share their 3D views and get a full 3D reconstruction of the environment as it goes through gradual changes.

The benefits of sharing the dynamic 3D view are clearly demonstrated in Fig. 2. There are two main explanations for the Azure Kinect cameras being incapable of capturing the monitors. First, these monitors have a reflective surface that can bounce off most of the IR illumination from the cameras. Thus, only a fraction of the signal can be received by the sensor. Second, the long distance between the camera and the monitors, as well as the large incident angle of the IR ray and the surface normal, contribute to the depth error. As the dynamic 3D sensor is free from the limitations of the static cameras, the lost information can be picked up by viewing at a closer distance or alternative angle.



Figure 3: (a) The drifting issue can be observed from multiple frames of point cloud directly accumulated together. (b) The corrected pose of each frame reveals a non-overlapping marker pattern.

The drifting problem of current HMDs' inside-out tracking systems is well-known in the AR/VR and SLAM community. As a result, the transformation cT_w presented in Equation 1 needs to be re-estimated at each time when the HMD world origin shifts. An interesting research direction would be to utilize the higher-quality live 3D reconstruction of the scene from the static 3D capturing system as ground truth mapping for the HMD. Therefore at regular intervals, we can register the point cloud to the known map and compute the pose of HMD to correct the drifting issue. We demonstrate this effect in a post-processing step shown in Fig. 3.

In conclusion, we here proposed the concept of Dynamic 3D View Sharing. In a telepresence scenario, our markerless registration approach allows the local user's HMD to be quickly calibrated to the 3D capturing system. We showcased the use and explored its benefits in a medical scenario. Thanks to the additional perspective from the HMD user, objects that are not being captured by the static RGBD cameras can be recovered. In addition, we discussed the potential research directions expanding the presented concept to dynamically improve the 3D world reconstruction and consequently users' mixed reality telepresence experience.

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